LECTURE 6

Def'n A <u>linear equation</u> is an equation of the form $a_1 x_1 + a_2 x_2 + \dots + a_n x_n = b$ where $a_i, b \in \mathbb{F}$ are known and x_i is unknown

Ex $2x_1 - x_2 + 5x_3 = 4$ is linear but $x_1 x_2 = 4$ is not

$\sin(x_1) + \sqrt{x_3} - x_2 = 0$

Def'n A <u>system of linear equations</u> is a system of equations of the form

$$\left.\begin{aligned} a_{11} x_1 + a_{12} x_2 + \dots + a_{1n} x_n &= b_1 \\ a_{21} x_1 + a_{22} x_2 + \dots + a_{2n} x_n &= b_2 \\ a_{m1} x_1 + a_{m2} x_2 + \dots + a_{mn} x_n &= b_m \end{aligned}\right\} \begin{array}{l} m \text{ equations} \\ \text{"}m \times n \text{ system"} \end{array}$$

Ex $x_1 + 2x_2 + x_3 = 2 \Rightarrow 1 \times 3$ system

A sol'n is $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$. Another is $\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$.

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 2 \\ x_1 \qquad - x_3 &= -4 \end{aligned} \Rightarrow 2 \times 3 \text{ system}$$

$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ is not a sol'n to the system because it doesn't hold for ②.

But $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 3 \\ 0 \end{bmatrix}$ is

There are infinitely many sol'ns. Indeed they are given by $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 + t \\ 3 - t \\ t \end{bmatrix}$ into our system.

linear equation, system of linear equations

Ex

Solve $\begin{cases} x_1 + x_2 + x_3 = 6 \\ 2x_1 - 2x_2 + x_3 = 1 \\ x_1 + x_2 - x_3 = 0 \end{cases} \Rightarrow 3 \times 3$ system

$$\begin{aligned} x_1 + x_2 + x_3 &= 6 && (e_1) \\ 2x_1 - 2x_2 + x_3 &= 1 && (e_2) \\ x_1 + x_2 - x_3 &= 0 && (e_3) \end{aligned}$$

$$\begin{aligned} x_1 + x_2 + x_3 &= 6 && (e_1) \\ 2x_1 - 2x_2 + x_3 &= 1 && (e_2) \\ -2x_3 &= -6 && (e_3) \longleftarrow (e_3) - (e_1) \end{aligned}$$

$$\begin{aligned} x_1 + x_2 + x_3 &= 6 && (e_1) \\ 2x_1 - 2x_2 + x_3 &= 1 && (e_2) \\ x_3 &= 3 && (e_3) \longleftarrow (e_3) \cdot \left(\frac{1}{-2}\right) \end{aligned}$$

$$\begin{aligned} x_1 + x_2 &= 3 && (e_1) \longleftarrow (e_1) - (e_3) \\ 2x_1 - 2x_2 &= -2 && (e_2) \longleftarrow (e_2) - (e_3) \\ x_3 &= 3 && (e_3) \end{aligned}$$

$$\begin{aligned} x_1 + x_2 &= 3 && (e_1) \\ -4x_2 &= -8 && (e_2) \longleftarrow (e_2) - 2 \cdot (e_1) \\ x_3 &= 3 && (e_3) \end{aligned}$$

$$\begin{aligned} x_1 + x_2 &= 3 && (e_1) \\ x_2 &= 2 && (e_2) \longleftarrow (e_2) \cdot \frac{1}{3} \\ x_3 &= 3 && (e_3) \end{aligned}$$

$$\begin{aligned} x_1 &= 1 && (e_1) \longleftarrow (e_1) - (e_2) \\ x_2 &= 2 && (e_2) \\ x_3 &= 3 && (e_3) \end{aligned}$$